

Ghilmo Choi

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Education

Korea Aerospace University , Republic of Korea B.S. in Mechanical Engineering	Mar 2021 – Feb 2027 (expected)
◦ Official GPA: 4.00 / 4.5 (Major GPA: 4.08 / 4.5)	
◦ Expected GPA after current-semester grades are posted: 4.08 / 4.5 (Major GPA: 4.18 / 4.5)	
◦ Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics	

Research Interests

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- On-device vision-language models for natural-language robotic mission execution
 - Perception, Prediction, and Decision Making for Autonomous Driving
 - Motion planning and path tracking for real-vehicle autonomous driving

Research / Project Experience

ASOL (Autonomous Systems and Optimization Lab) - <i>Undergraduate Researcher</i>	Korea Aerospace Univ. June 2025 – Present
◦ On-device VLM-based robotic mission execution Developed a natural-language mission execution framework that converts user commands into executable robot behaviors on a UGV platform	
◦ Hallucination-aware and SLAM-grounded task planning Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution	
◦ AirSim-based perception, reconstruction, and path generation Extending the framework with YOLOE, NanoOWL, and Grounding DINO for object detection on AirSim data, NVBlox for 3D reconstruction, and OMPL for path generation	
KAUVOY Autonomous Driving Club - <i>Localization Part</i>	Korea Aerospace Univ. Nov 2024 – Nov 2025
◦ EKF-based localization Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements	
◦ Autonomous parking and path tracking Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid A* path planning, and Pure Pursuit-based path tracking	
- <i>Team Leader</i>	Nov 2025 – present
◦ Finite-State-Machine-based decision-making module for autonomous driving scenarios	
Course Project, Introduction to Machine Learning - End-to-End Machine Learning Classification and Regression Pipeline	Korea Aerospace Univ. Aug 2025 – Nov 2025
◦ Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved 0.915 macro F1 on a 4-class imbalanced classification task	

Honors & Activities

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- Finalist, **2025 College Mobility Innovation Competition**
 - Excellence Award, **Drone X AI Marketing Challenge** (2026)
 - 2026 - **Airbus 101 Competition** (5th Place)

Publications

- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS `robot_localization` Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

Technical Skills

- Python, C/C++, MATLAB, Linux(Ubuntu), ROS1, ROS2, Docker, Git